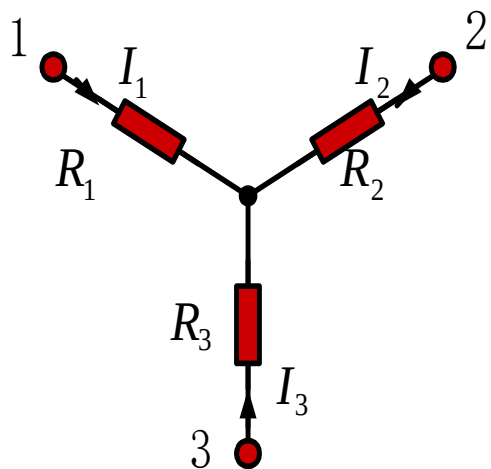


第二章 线性直流电路

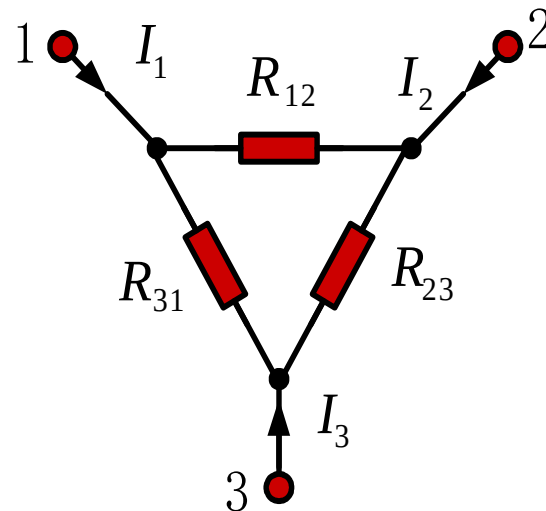


2.1 电阻网络的等效

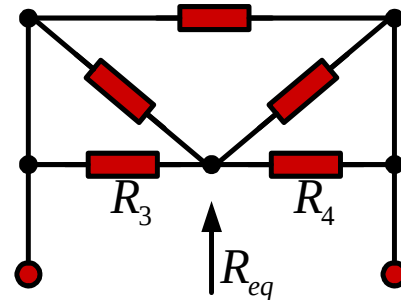
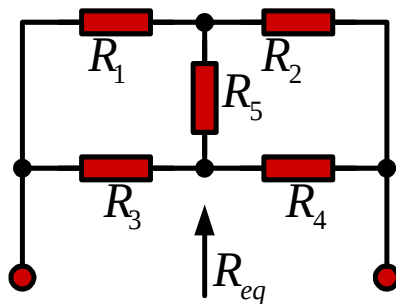
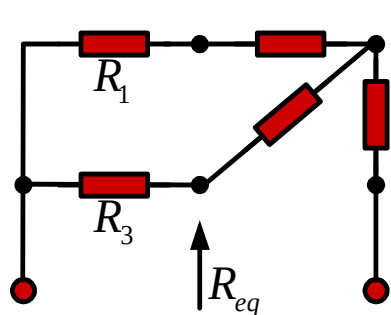
4. 星形联结和三角形联结



星形(T形)联接

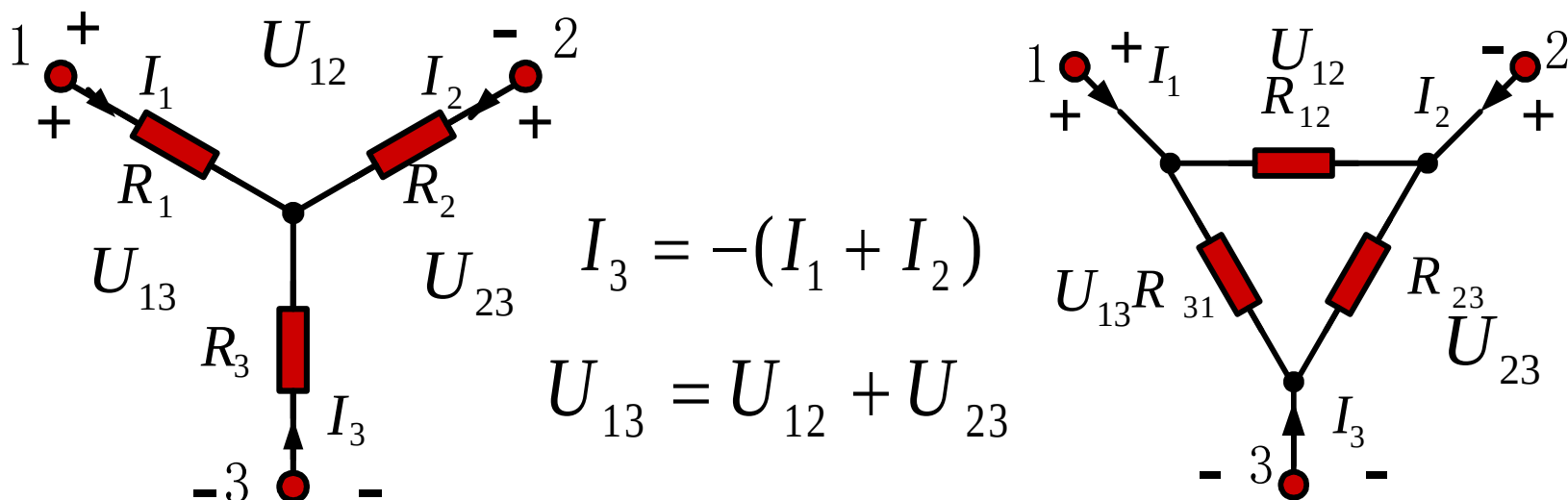


三角形(Δ 形)联接



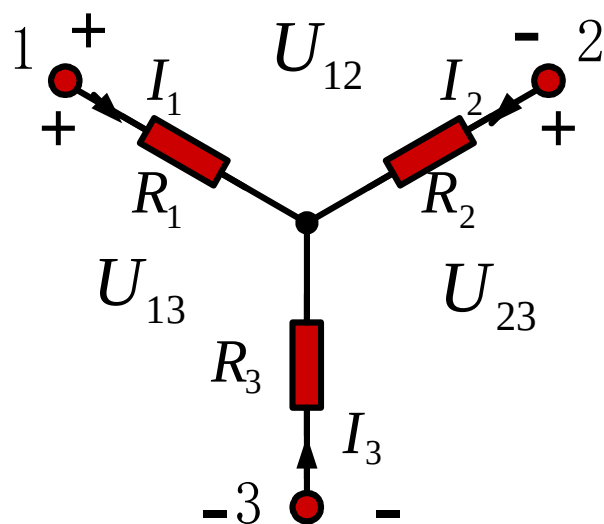
2.1 电阻网络的等效

等效条件:



分析：将星形连接转换成三角形连接时，将减少一个节点，但要增加一个回路；而将三角形连接转换成星形连接时，将减少一个回路，但要增加一个节点。

2.1 电阻网络的等效

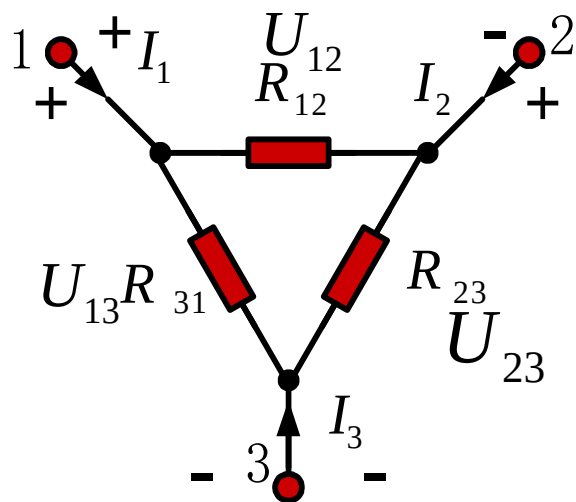


• 星形连接中的电压、电流关系

$$\begin{bmatrix} U_{13} \\ U_{23} \end{bmatrix} = \begin{bmatrix} R_1 + R_3 & R_3 \\ R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\mathbf{U} = \mathbf{R}\mathbf{I}$$

• 三角形连接中的电压、电流关系

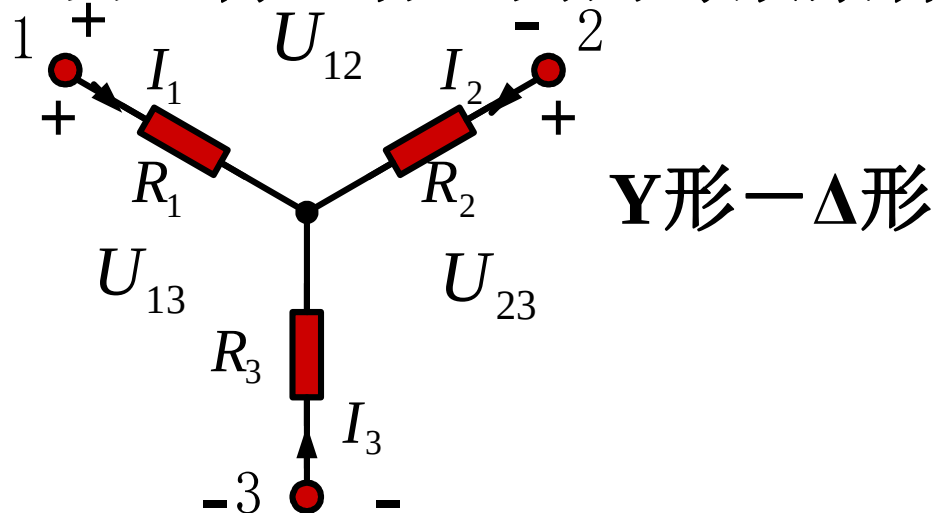


$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} G_{12} + G_{31} & -G_{12} \\ -G_{12} & G_{12} + G_{23} \end{bmatrix} \begin{bmatrix} U_{13} \\ U_{23} \end{bmatrix}$$

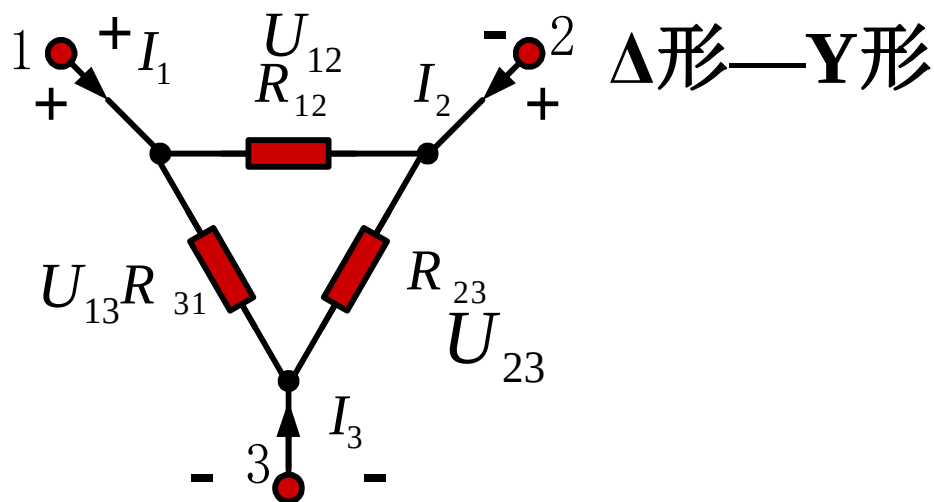
$$\mathbf{I} = \mathbf{G}\mathbf{U} \quad \mathbf{U} = \mathbf{G}^{-1}\mathbf{I}$$

2.1 电阻网络的等效

• 由此得二者之间的等效条件是



$$\left. \begin{aligned} R_{12} &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \\ R_{23} &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \\ R_{31} &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \end{aligned} \right\}$$



$$\left. \begin{aligned} R_1 &= \frac{R_{12} R_{31}}{R_{12} + R_{23} + R_{31}} \\ R_2 &= \frac{R_{23} R_{12}}{R_{12} + R_{23} + R_{31}} \\ R_3 &= \frac{R_{31} R_{23}}{R_{12} + R_{23} + R_{31}} \end{aligned} \right\}$$

2.1 电阻网络的等效

三个相等的电阻接成Y形或Δ形时的等效变换是：

$$\begin{aligned} R_1 &= R_2 = R_3 = R_Y \\ R_{12} &= R_{23} = R_{31} = R_{\Delta} = 3R_Y \end{aligned} \quad \Rightarrow \quad R_Y = \frac{1}{3} R_{\Delta}$$

Y形—Δ形

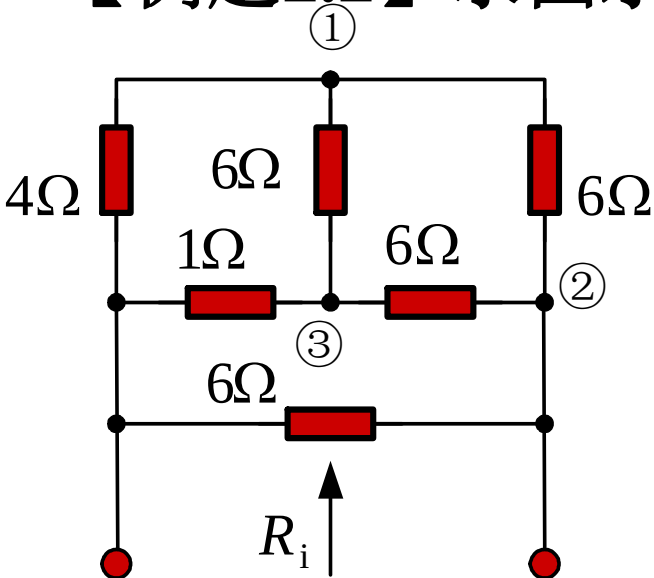
$$\left. \begin{aligned} R_{12} &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \\ R_{23} &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \\ R_{31} &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \end{aligned} \right\}$$

Δ形—Y形

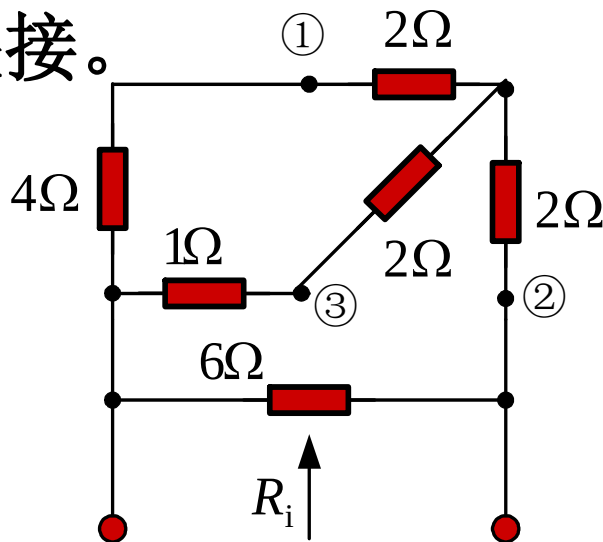
$$\left. \begin{aligned} R_1 &= \frac{R_{12} R_{31}}{R_{12} + R_{23} + R_{31}} \\ R_2 &= \frac{R_{23} R_{12}}{R_{12} + R_{23} + R_{31}} \\ R_3 &= \frac{R_{31} R_{23}}{R_{12} + R_{23} + R_{31}} \end{aligned} \right\}$$

2.1 电阻网络的等效

【例题2.2】求图示电路的等效电阻 R_i ？

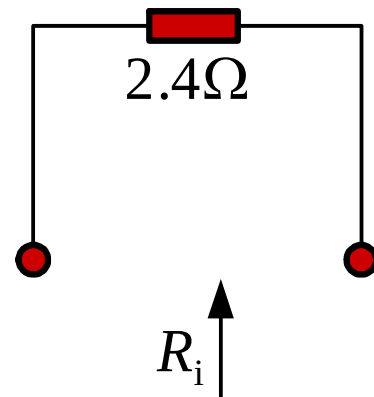


解：将节点①、②、③之间的对称 Δ 形联接电阻化为等效对称的Y形联接。



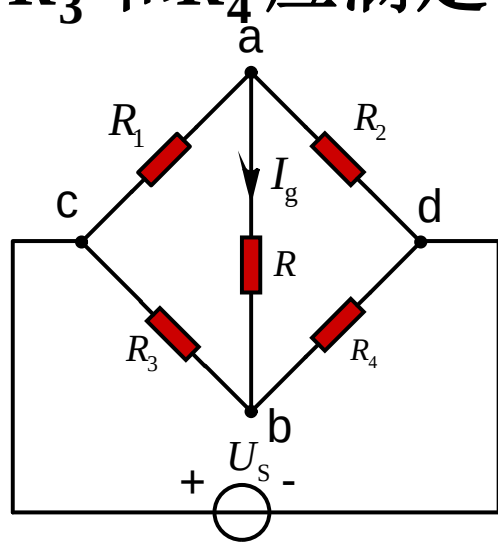
用串并联化简等效后的电路求出等效电阻

$$R_i = 6\Omega \parallel [(4\Omega + 2\Omega) \parallel (1\Omega + 2\Omega) + 2\Omega] = 2.4\Omega$$

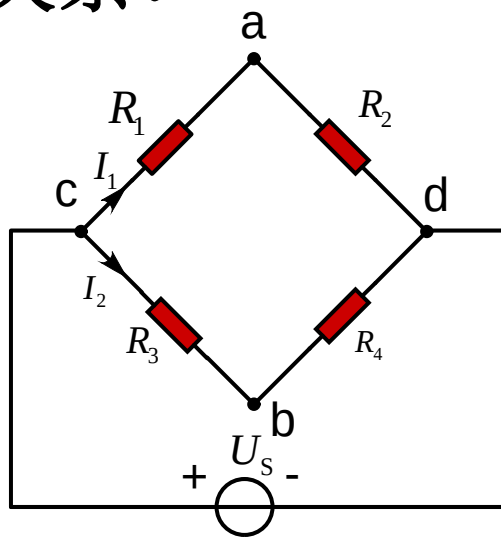


2.1 电阻网络的等效

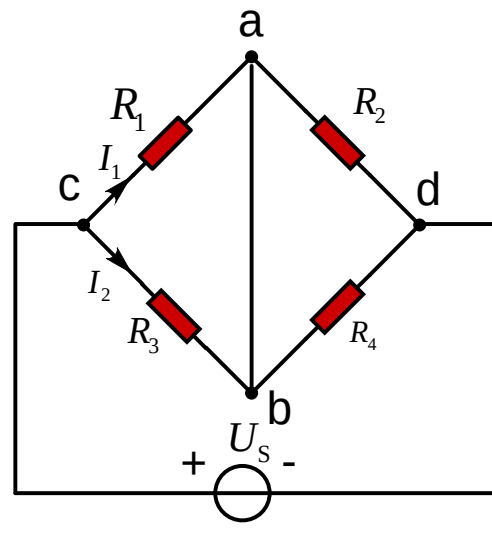
【例题2.3】图示电路是桥形电阻电路，当 $I_g=0$ ， $U_{ab}=0$ 时称电桥是平衡的，试说明电桥平衡时的电阻 R_1 ， R_2 ， R_3 和 R_4 应满足什么关系？



(a)



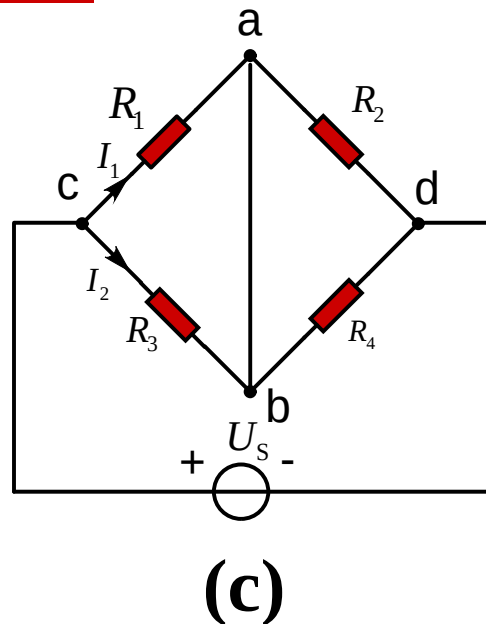
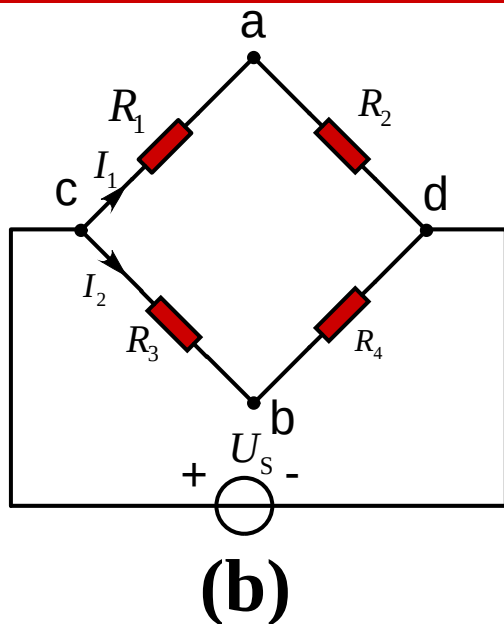
(b)



(c)

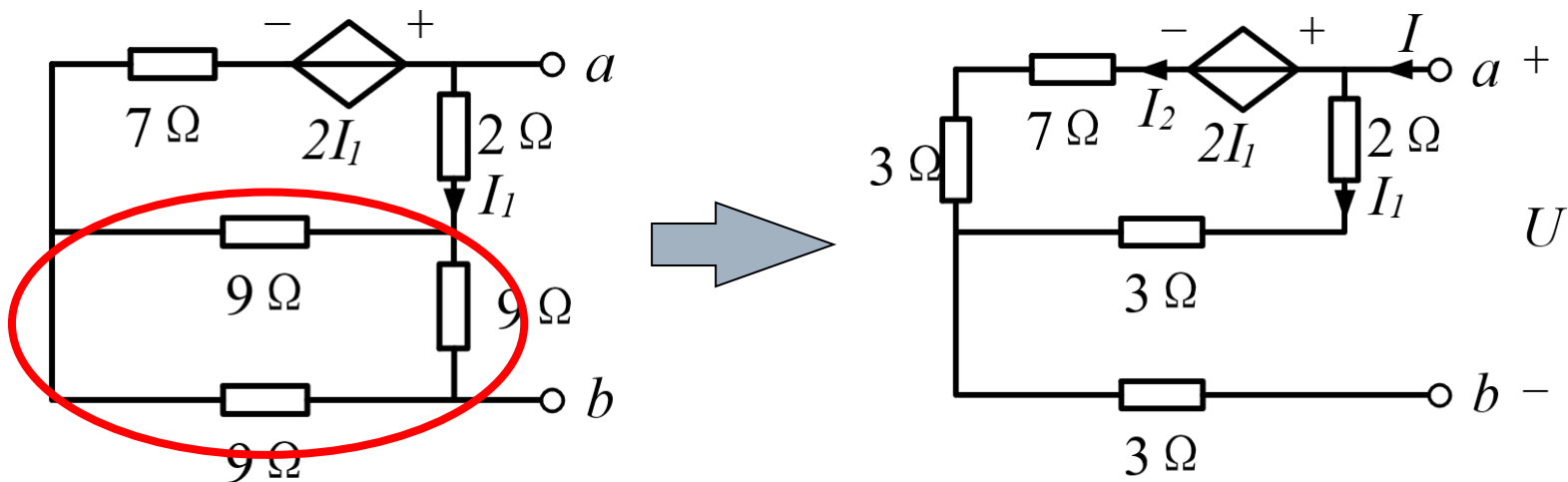
解：电桥平衡时 $I_g=0$ ， $U_{ab}=0$ 。从 $I_g=0$ 角度看， a ， b 两点间是开路的，如图(b)所示；从 $U_{ab}=0$ 角度看， a ， b 两点间是短路的，如图(c)所示。

2.1 电阻网络的等效



$$\begin{aligned} \text{由图b有 } I_1 &= \frac{U_s}{R_1 + R_2} \quad I_2 = \frac{U_s}{R_3 + R_4} \\ \text{由图c有 } U_{ca} &= U_{cb} \quad \text{即 } R_1 I_1 = R_3 I_2 \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{由图b有 } I_1 &= \frac{U_s}{R_1 + R_2} \quad I_2 = \frac{U_s}{R_3 + R_4} \\ \text{由图c有 } U_{ca} &= U_{cb} \quad \text{即 } R_1 I_1 = R_3 I_2 \end{aligned}} \right\} \begin{aligned} \frac{R_1 U_s}{R_1 + R_2} &= \frac{R_3 U_s}{R_3 + R_4} \\ \text{即 } \frac{R_1}{R_2} &= \frac{R_3}{R_4} \end{aligned}$$

【例题a】求如图所示电路的等效端口电阻。



解：角形连接转换为星型连接

$$I_2 = \frac{-2I_1 + (3+2)I_1}{3+7} = 0.3I_1$$

$$I = I_1 + I_2 = 1.3I_1$$

$$U = (2+3)I_1 + 3I = 8.9I_1$$

$$R = \frac{U}{I} = \frac{8.9I_1}{1.3I_1} = \frac{89}{13} \Omega$$

谢

谢！